Actual twist transfer over $\mathbf{F}_{37}$

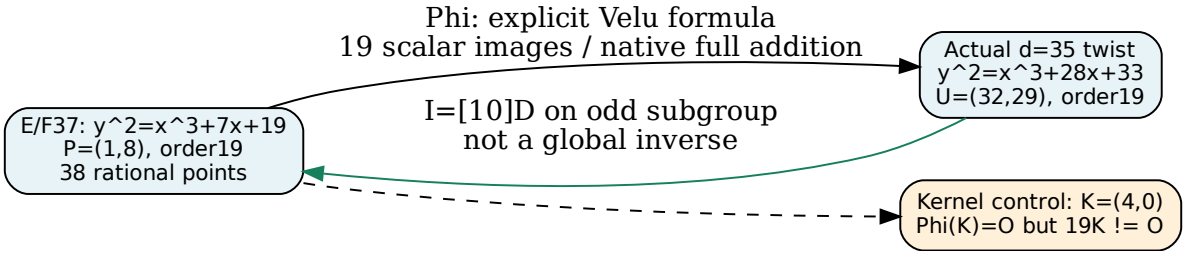
2026-10-30 PR2381 fixed public synthetic fixture

Let $E : y^2 = x^3 + 7x + 19$ and $T : y^2 = x^3 + 28x + 33$, the actual monic twist with nonsquare $d = 35$. Each has 38 rational points. The fixed lexicographic rule selects $P = (1, 8)$ and $U = (32, 29)$, both of order 19. These are unregistered model tuples; no endomorphism-ring classification is inferred.

$$\Phi(x, y) = \left(x + \frac{18}{x - 4},\ y\left(1 - \frac{18}{(x-4)^2}\right)\right),$$

$$D(x, y) = \left(28\left(x + \frac{35}{x - 29}\right),\ 14y\left(1 - \frac{35}{(x-29)^2}\right)\right).$$

Origins and the respective kernel points $(4, 0)$ and $(29, 0)$ map to the origin. The inverse on the order-19 subgroups is $I = [10]D$.



Independent native verification. All $38 + 38$ point images, both compositions $= [2]$, 2888 addition pairs, and 19 scalar images/inverses pass. The false inverse $[9]D$ differs on all 18 nonzero subgroup points.

Lean result. The unmodified scoped gate passes 19 public named theorems and 22 audited declarations, with zero failing or extra trusted/native axioms. The actual Mathlib nonsingular points use the displayed coordinate formulas. Lean checks equations, order, scalar transport, subgroup addition, inverse identities, and a necessary coefficient obstruction to a short-model change. For an actual unknown source point Q and $n \in \mathrm{Fin}\,19$:

$$(\exists i \in \mathrm{Fin}\,19,\ Q = iP) \implies [\Phi(Q) = nU \iff Q = nP].$$

This does not give a solver the unknown index. Global inversion is false at the source 2-torsion point. Generic geometric classification is not formalized.

Full size-two policy frontier

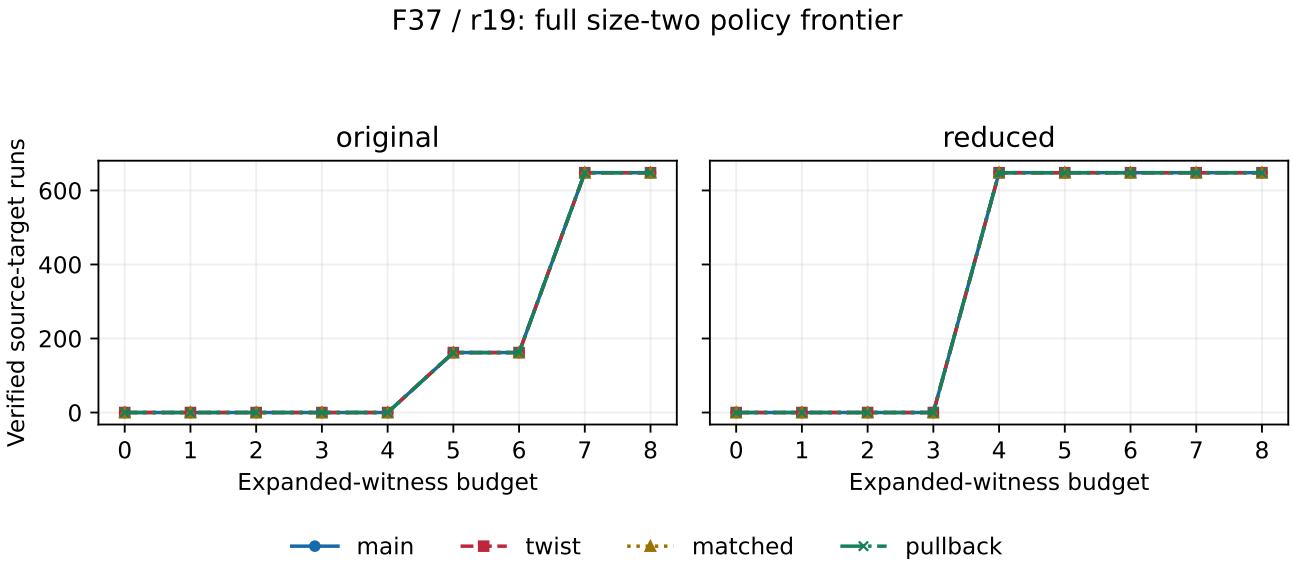


Figure. Native-emitted counts for 36 policies and 18 source points per role and cap. Curves may overlap. These are exact finite values; no timing or sampling uncertainty is inferred.

All 144 arms, including matched and pullback controls, run both encodings at caps 0–8: 2592 configurations and 46656 separate target attempts. The latter repeat 18 source points across policies, encodings and budgets.

Independently reconstructed outcome	Count
Verified original-source recoveries	19440
Blocked public-log setups	1512
Policy/target/budget pair records	3779136
Higher rank, lower requested cap, recovery admitted	55230
Such records beating the strongest source policy	0
Changed controls rejected, no receipt	10
Shared backend tests passed	22

The 55230 records are pair comparisons, not distinct target wins. A lower requested cap need not mean a lower actual solving degree. None of these gains survives comparison with the strongest canonical source policy at the comparison budget. Stationary yield, intrinsic regularity, runtime and complete-cost gains remain unestablished. The smaller F13 cohort is not a matched speed reference.

Execution, evidence and next test

Algebraic instrument. New F37/r19 sources receive three inputs: S_3 and both product-membership constraints. Flat Macaulay rows carry exact original-input witnesses. A unit or bounded Buchberger certificate must pass before root extraction. Both matrix degree and expanded verification-witness degree fit the rung. Above-bound shifts are denied before construction; an unknown cap result admits no roots or rows. A smaller already-computed witness is retained only for the identical polynomial within one solve.

Public setup derives logs only from admitted relations and checks the actual base points. Incomplete rank blocks every target. Recovery receives actual unknown Q, makes fresh shifted solves, retains every failed or identity shift and sign lift, and verifies the scalar on the original E/P. Output-reference binding follows each fresh solve and is unpriced. Direct enumeration is a separate generic decomposition reference, with no named matrix degree. Setup, allocation, field/group operations and verification lack a complete price.

Preservation and failures. Complete producer and independent auditor paths pass. Nine census corruptions and one false-inverse payload genuinely change and reject with exit 101 and no receipt. All canonical cases, altered payloads, control reconstruction, raw commands/logs and proof receipts remain. The 18 prior polynomial identity records are unchanged; no new identity or combined/global proof audit is claimed.

The first Lean setup failed before elaboration while cloning Mathlib. An exact pinned dependency cache was then reused without changing the gate. Attempt 02 has an explicit successful gate exit and PASS audit despite an outer PTY exit discrepancy. An unchanged-source repeat explicitly returned zero/PASS; both records remain. A copied F13-ordering phrase in the frozen protocol has a dated metadata erratum. The new F37 scope was already explicit in the question, arms, parameters, Markdown, sources and coverage assertions. No criterion changed.

Next test. Use all size-three bases over this verified route: 84 canonical policies per curve plus matched and pullback controls. A new protocol and instrument are required. Since S_3 has degree at most two per variable, cubic membership reduction leaves it unchanged; this must be an explicit encoding-equality control. This next experiment is not implemented or run here.

The PR stays open, non-draft, with auto-merge off. Registry, incomplete library suite, naming, required CI, review, integration and Conductor holds remain. The scientific window was 13:40–14:40 UTC, with two local workers and 600-second command caps. No paid resource or arbitrary result upload was used.

Primary source: Sage isogeny blob 972f4116…, Kohel/Vélu formulas and sum helper. Full source links, proof map, hashes and replay instructions are in the adjacent README and validation directory.